Conditional Independence Test with Tolerance

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1 Introduction

Suppose we want to know whether a random variable X is related to Y after controlling for a set of confounding variables Z. One way to characterize this problem is formalize as a conditional independence test:

$$H_0 : X \perp\!\!\!\perp Y \mid Z \qquad H_1 : X \not\perp\!\!\!\perp Y \mid Z$$

Equivalently, it can be inscribed by conditional mutual information:

$$X \perp\!\!\!\perp Y \mid Z \quad \Longleftrightarrow \quad I(X;Y|Z) = 0$$

where $I(X;Y|Z) = \mathbb{E}_z\big[D_{KL}[\,p(x,y\,|z)\,||\,p(x|z)p(y|z)\,]\big]$.

To test for conditional independence, we can test whether the conditional mutual information is equal to 0. However, in the real data scenario, many variables are usually weakly related and $I(X;Y|Z)$ is not exactly zero, leading us to find variables that are not particularly important and have little explanatory power. To avoid finding too many weak signals, a natural idea is to add some tolerance to the test: we consider X to have sufficient explanatory power only if $I(X;Y|Z)$ exceeds some pre-specified threshold τ. Then, our testing problem becomes:

$$H_0 : I(X;Y|Z) \leq \tau \qquad H_1 : I(X;Y|Z) > \tau$$

Directly estimate $I(X;Y|Z)$ from finite sample is a quite hard problem. Many existing methods for conditional independence testing do not involve explicit estimation and instead using randomization test by resampling [1] or permutation [2] to generate conditional independent samples $\tilde{x}^{(m)}$. They choose a score statistic $T(x,y,z)$, and compare the original one with its counterpart $T(\tilde{x}^{(m)},y,z)$. If $X \perp\!\!\!\perp Y \mid Z$, then $T(x,y,z)$ and $T(\tilde{x}^{(m)},y,z)$ are conditionally i.i.d. and exchangeable so that the rank of $T(x,y,z)$ will be uniform. Hence we can transform the problem of testing for conditional independence into testing whether the rank is uniform, by testing its tail probability [1][2] or uniformity using chi-square statistic [3]. These methods rely on knowledge of the conditional distribution $p(x|z)$, which is known as model-X setting.

As mentioned above, the validity of such test relies on the exchangeability of score statistics $T(x,y,z)$ and $T(\tilde{x}^{(m)},y,z)$. However, when H_0 becomes

$I(X;Y|Z) \leq \tau$, exchangablity may not hold because we allow true distribution can be a little different from conditional independence. To address this issue, we adopt the approach in [3] which change the testing target from uniform to multinomial distribution to allow the non-equal probabilities under H_0 by limiting their KL-divergence to some degree.

2 Method

Notations Use $\mathcal{D} = \{(X_1, Y_1, Z_1), ..., (X_n, Y_n, Z_n)\} \in \mathcal{X}^n \times \mathcal{Y}^n \times \mathcal{Z}^n$ to denote n i.i.d. samples from $\mathcal{P}_0$ with density $p(x, y, z)$. We use $(X_i^{(m)}, Y_i, Z_i)$, $X_i^{(m)} \sim p(x|z = Z_i)$ to denote new generated samples. $T_i = T(X_i, Y_i, Z_i)$ is score function ($T_i^{(m)}$ by same argument). R_i is rank of T_i among $(T_i, T_i^{(1)}, ..., T_i^{(B)})$. We devide $\{1, 2, ..., B + 1\}$ into equal L segments for some right K satisfying $KL = B + 1$. And $V_l = \sum_{i=1}^{n} \mathbf{1}\{K(l - 1) < R_i \leq Kl\}$ denotes observation frequency on lth segments. We use Multi$(p_1, ..., p_L)$ and Uniform(L) to represent multinomial and uniform distribution with corresponding parameter, respectively.

Algorithm 1 χ^2-randomization test with tolerance

Input: Dataset $\mathcal{D} = \{(X_1, Y_1, Z_1), ..., (X_n, Y_n, Z_n)\}$, a score function $T(x, y, z)$ chosen independent of $\mathcal{D}$, resampling size B.

Output: p-value

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for i in 1, 2, ..., n do
    Compute T_i = T(X_i, Y_i, Z_i).
    for m in 1, 2, ..., B do
        Sample X_i^(m) from p(x|z = Z_i).
        Compute T_i^(m) = T(X_i^(m), Y_i, Z_i).
    end for
    Denote R_i the rank of T_i among {T_i, T_i^(1), ..., T_i^(B)}.
end for
for l in 1, 2, ..., L do
    Compute observation frequency V_l = Σ_{i=1}^n 1{K(l−1) < R_i ≤ Kl} of lth
    segment.
end for
Solve the optimization problem (1) to get statistics U.
p = 1 − F^{-1}_{χ²_{L−1}}(U) where F_{χ²_{L−1}} is cdf of χ² distribution with degree of
    freedom L − 1.
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When $0 < I(X;Y|Z) < \tau$, $T(x, y, z)$ and $T(\tilde{x}^{(m)}, y, z)$ are no longer exchangeable and R_i is generally not uniform. But it will not too far from uniform because of the limitation τ. [4] characterize its maximum deviance in f-divergence fashion and we state it in Proposition 1 (in KL-divergence form).

Proposition 1. $D_{KL}[\, Multi(p_1, ..., p_L) \,||\, Uniform(L)\,] \leq I(X;Y|Z)$

Therefore, we can use χ^2 statistics to test if rank R_i follows $\text{Multi}(p_1, ..., p_L)$. The only problem is that we do not know true probability. But [4] suggest that we can bypass this obstacle by writing it as an optimization problem:

$$\begin{aligned} U = \min_{p_1,...,p_L} \sum_{l=1}^{L} \frac{(V_l - np_l)^2}{np_l} \\ s.t. \sum_{l=1}^{L} p_l = 1, \qquad p_l \geq 0, \qquad \sum_{l=1}^{L} p_l \log(Lp_l) \leq \tau \end{aligned} \tag{1}$$

Note that $D_{KL}[\,\text{Multi}(p_1, ..., p_L) \,||\, \text{Uniform}(L)\,] = \sum_{l=1}^{L} p_l \log(Lp_l)$. We summarize details in Algorithm 1.

Theorem 1. *Assume that $H_0 : I(X;Y|Z) < \tau$ is true, and conditional density $p(x|z)$ is given, then for any function $T : \mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \to \mathbb{R}$, the p-value given by Algorithm 1 has asymptotic size no more than α, i.e. $lim_{n\to\infty} \mathbb{P}(p_n \leq \alpha) \leq \alpha$.*

Theorem 1 tells us our p-value is asymptotic valid under H_0, and its validity holds for all function T. However, the power of test depend on the choice of T so we should carefully choose right statistics to differentiate null and alternatives. Some common statistics are partial correlation coefficient, absolute value of lasso coefficient etc. Further, we can leverage machine learning model to learn a prediction model $\hat{f}$, then choose test statistics as $T = -\mathcal{L}(\hat{f}(X, Z), Y)$ for some loss function $\mathcal{L}$. If $I(X;Y|Z) \neq 0$, T will stochastic larger than uniform for sufficiently good model. The only caveat is that $\hat{f}$ should be independent of $\mathcal{D}$ thus must be given or estimated from other dataset.

3 Simulation

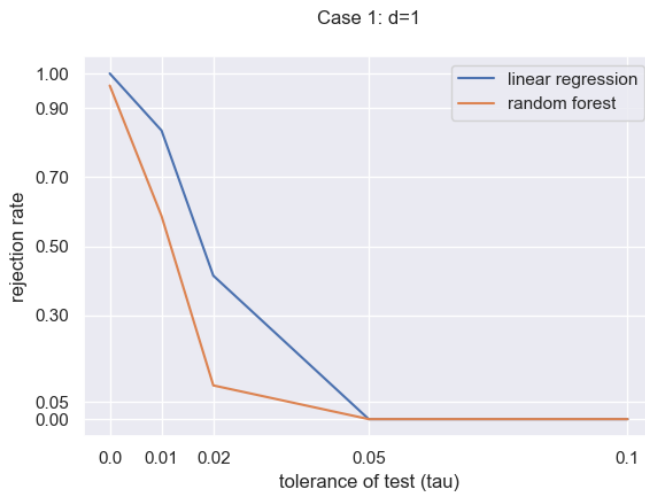
We use score statistics in the form of $T = -\mathcal{L}(\hat{f}(X, Z), Y)$, with $\mathcal{L}$ is L_2 loss and $\hat{f}$ to be learned from linear regression, random forest or lasso.

We fix dimension of X and Y to 1 and set dimension of Z as $d = 1$ or $d = 50$. All variables are gaussian: $(X, Z) \sim N(0, \Sigma_{d+1}), \Sigma_{ii} = 1, \Sigma_{ij} = \rho = 0.5$ for $i \neq j$, and use two different regression model:

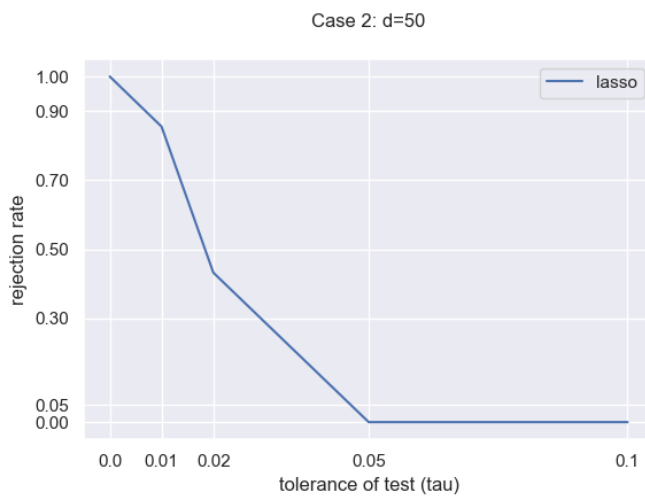
(1)$Y = cX + \beta Z + \epsilon$ with $d = 1$, $\beta \sim N(0, 2)$.

(2)$Y = cX + \beta^T Z + \epsilon$ with $d = 50$, $\beta_i \sim N(0, 2)$ for $1 \leq i \leq 5$ and $\beta_i = 0$ for $i > 5$.

When $c = 0$, Y is conditionally independent with X on Z, and c is the magnitude of dependence controlled by $I(X;Y|Z)$ with relation $c^2 = (e^{2I(X;Y|Z)} - 1)(1 + \rho(d-1)) \,/\, ((1-\rho)(1+\rho d))$. We fix the value of $I(X;Y|Z)$ as 0.1 letting X and Y have weak relations, and vary tolerance τ from 0 to $I(X;Y|Z)$. Note that when $\tau = I(X;Y|Z)$, H_0 holds thus it reflects size of test. Otherwise, it reflects the power of test. Figure 1 depict the rejection rate of such test averaged over 400 trails. Resampling size $B = 99$, and $K = 10$ to form $L = 10$ segments. The model $\hat{f}$ is learned by an independent training set with $n_{train} = n = 500$.



(a)



(b)

Figure 1: Rejection rate of χ^2 randomization test

When $d = 1$, we use linear regression and random forest. When $d = 50$, lasso is preferred.

Reference

[1] Candes, E., Fan, Y., Janson, L. and Lv, J. (2018) Panning for gold: 'model-X' knockoffs for high dimensional controlled variable selection. *J. R. Statist. Soc. B*, 80, 551–577.

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[3] Javanmard A., & Mehrabi M. (2021). Pearson chi-squared conditional randomization test. *arXiv:2111.00027*

[4] Javanmard A., & Mehrabi M. (2024). GRASP: a goodness-of-fit test for classification learning. *Journal of the Royal Statistical Society Series B (Statistical Methodology)*, 86, 215–245.